

BANA 698 Final Presentation

From Social Proof to Action

Algorithmic amplification as a conditional mechanism with reach saturation and authority effects

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Agenda

1**Research problem and gap****2****Research model and hypotheses****3****Data and methodology****4****Key findings****5****Contributions, limitations, and Takeaways**

Research Problem and Gap

Why separate visibility from meaningful engagement?

Business / research problem

Posts are not seen solely because users choose them. Platform ranking and recommendation systems allocate attention at scale.

Platforms allocate attention algorithmically, not neutrally.

Reach does not always convert into meaningful action.

Performance metrics can overstate true effectiveness.

Gap in prior work

Prior studies often treat reach and engagement as linked outcomes, but do not clearly separate early social proof, amplification, and high-effort action.

Low-effort and high-effort engagement are often blurred.

User response and algorithm response are rarely separated clearly.

Authority and saturation are under-modeled together.

Our contribution

This project tests whether early persuasive signals become action through amplification, and whether authority and high visibility change that pathway.

Separates signal, amplification, and outcome stages.

Tests authority as a boundary condition.

Shows diminishing returns under high visibility.

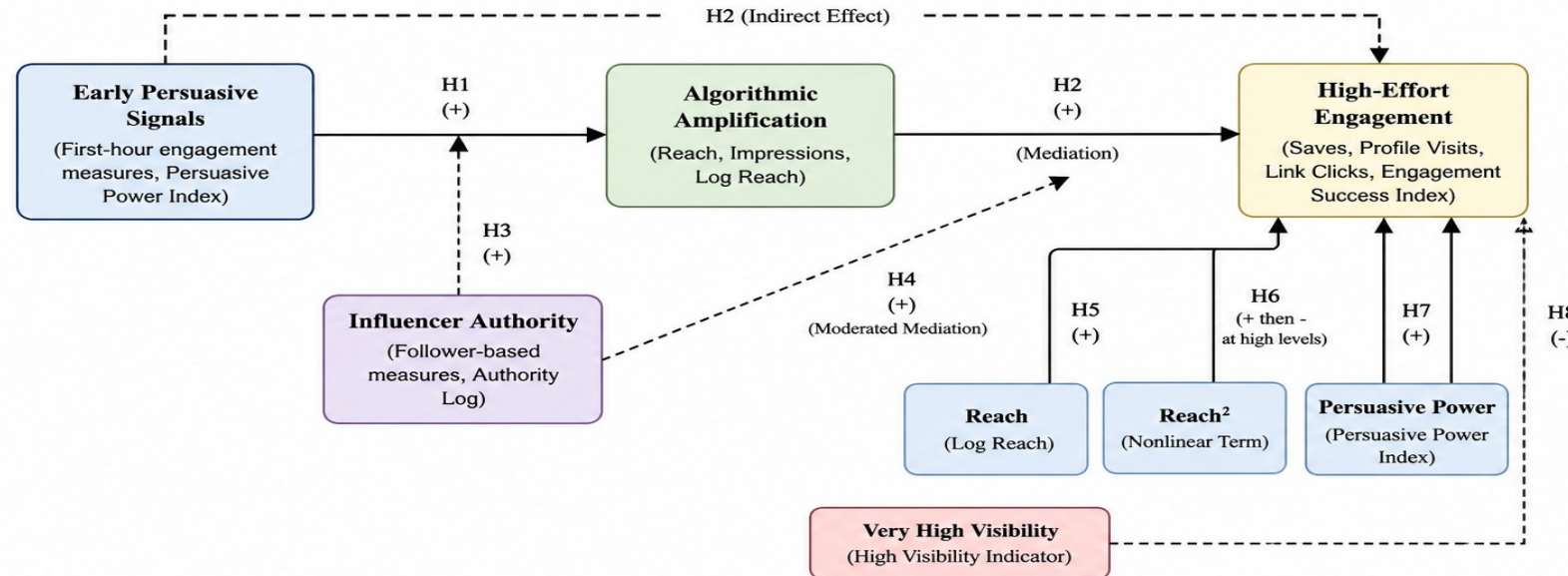
This study addresses the gap by examining 10,000 social media posts from 800 influencers, with a focus on how social proof becomes meaningful action under algorithmic visibility.

Core distinction: visibility is not the same as effectiveness.

Research Model and Hypotheses

Show the logic, not every detail

Figure 5.1. Conceptual Research Model



RQ1: Mechanism + authority

Do early signals translate to high-effort engagement through amplification, and does authority shape the path?

RQ2: Diminishing returns

At what point does additional reach add less meaningful engagement?

RQ3: High-visibility condition

Does very high visibility reduce the predictive role of persuasive power?

Legend

- Direct Effect
- Moderation
- > Mediation
- > Interaction / Boundary Condition

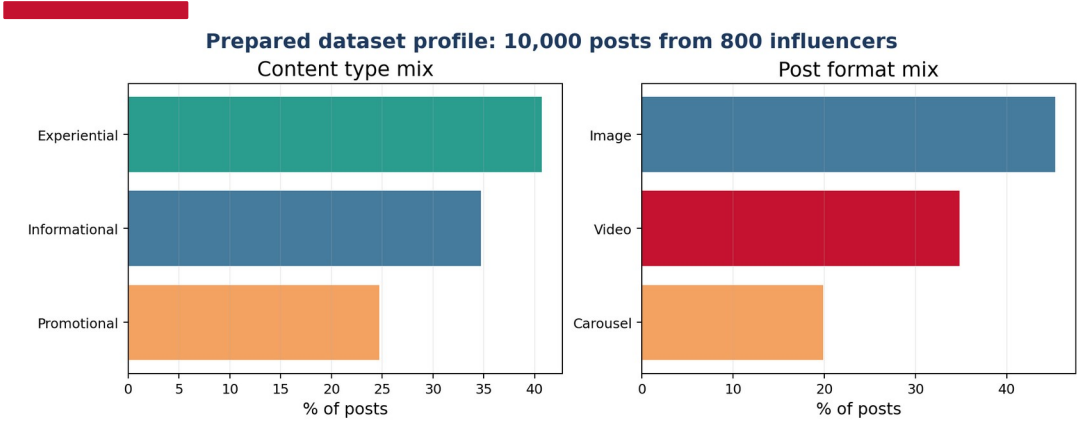
Notes on Hypotheses

- H1: Early persuasive signals → Algorithmic amplification (positive)
- H2: Algorithmic amplification mediates the relationship between early persuasive signals and high-effort engagement (positive indirect effect)
- H3: Influencer authority moderates the relationship between early persuasive signals and algorithmic amplification (stronger at higher authority)
- H4: The indirect effect of early persuasive signals on high-effort engagement through amplification is stronger for high-authority influencers
- H5: Reach → High-effort engagement (positive at lower to moderate visibility)
- H6: Reach exhibits diminishing marginal returns on high-effort engagement (positive association weakens at higher reach)
- H7: Persuasive power → High-effort engagement (positive at low to moderate visibility)
- H8: Very high visibility × Persuasive power → High-effort engagement (negative interaction; persuasive power less predictive at very high visibility)

Theory lens: HSM explains early cues; Source Credibility Theory explains why authority changes cue interpretation.

Data and Methodology

What data did we use and how did we analyze it?



Measures

IV: persuasive_power_index
Mediator: reach / impressions / log reach
DV: high-effort engagement
Moderator: authority_log

Methods

Regression-based mediation + moderation
OLS with clustered SE by influencer
Quadratic reach + high-visibility interaction

10,000

social media posts

800

influencers

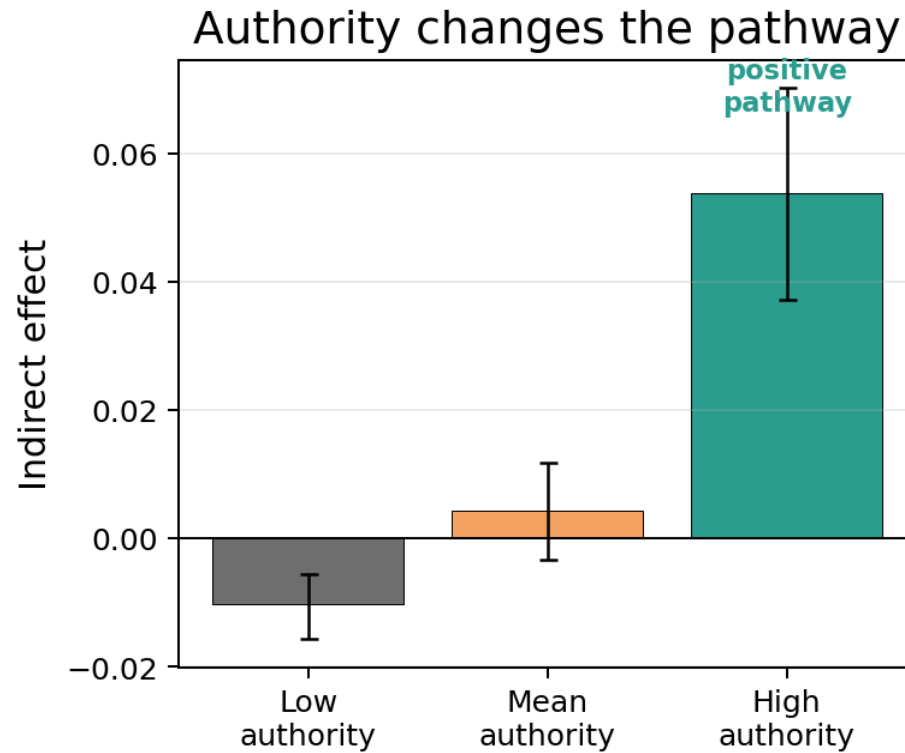
2024–2025

timestamped posts

Important limitation: prepared instructional dataset; algorithmic amplification is inferred through visibility proxies, not directly observed platform internals.

Key Findings I: Authority Changes the Pathway

RQ1 — mechanism and authority



Finding 1

H1 Supported
Early persuasive signals are positively associated with algorithmic amplification.

Finding 2

H2 Supported
Early persuasive signals influence high-effort engagement indirectly through amplification.

Finding 3

H3 and H4 Supported
Authority strengthens the pathway; the indirect effect is weakest at low authority and strongest at high authority.

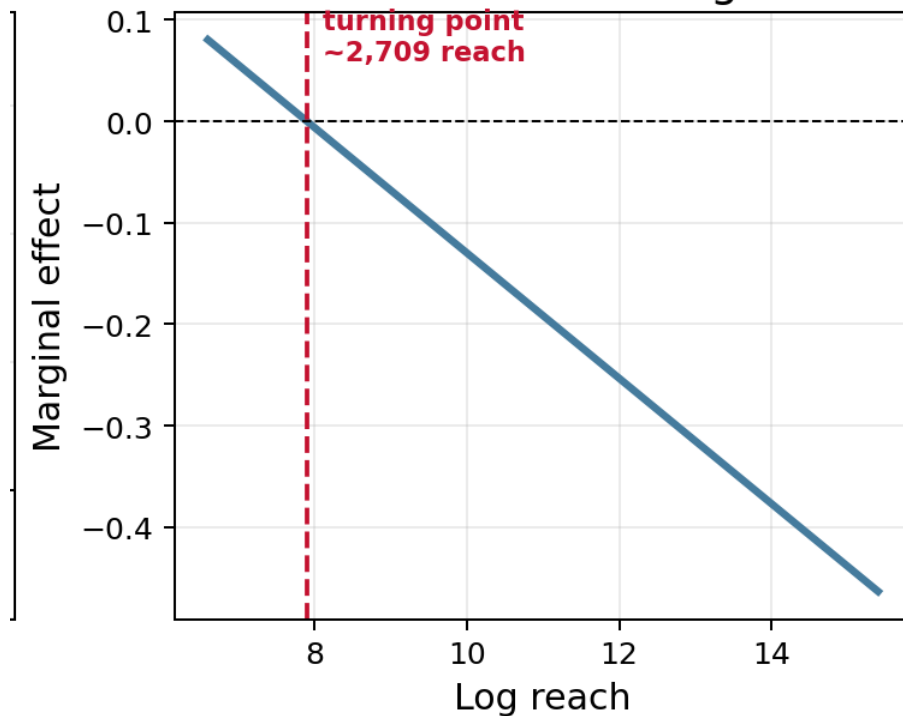
Bottom line: authority is the clearest boundary condition in the pathway from early signals to meaningful action.

Key Findings II: Visibility Has Limits

RQ2 and RQ3 — diminishing returns and very high visibility

Key empirical patterns

Reach shows diminishing returns



Finding 1

H5 Supported
Reach is positively associated with high-effort engagement at lower to moderate visibility levels.

Finding 2

H6 Supported
Reach shows diminishing marginal returns. Turning point: log reach = 7.90; actual reach ≈ 2,709.

Finding 3

H7 and H8 Supported
Persuasive power matters under lower/moderate visibility but becomes less predictive under very high visibility.

Bottom line: visibility helps, but beyond a point it adds less value and weakens persuasive differentiation.

Contributions, Limitations, and Takeaways

Academic contribution

Shows algorithmic amplification as a conditional mechanism, not just a descriptive outcome.

Separates signal, amplification, and outcome stages.

Shows authority and saturation as key boundary conditions.

Managerial contribution

Measure success with saves, clicks, and profile visits—not reach alone. Prioritize early quality and authority fit.

Helps identify when early momentum converts into action.

Shows that more reach is not always more value.

Limitations / next step

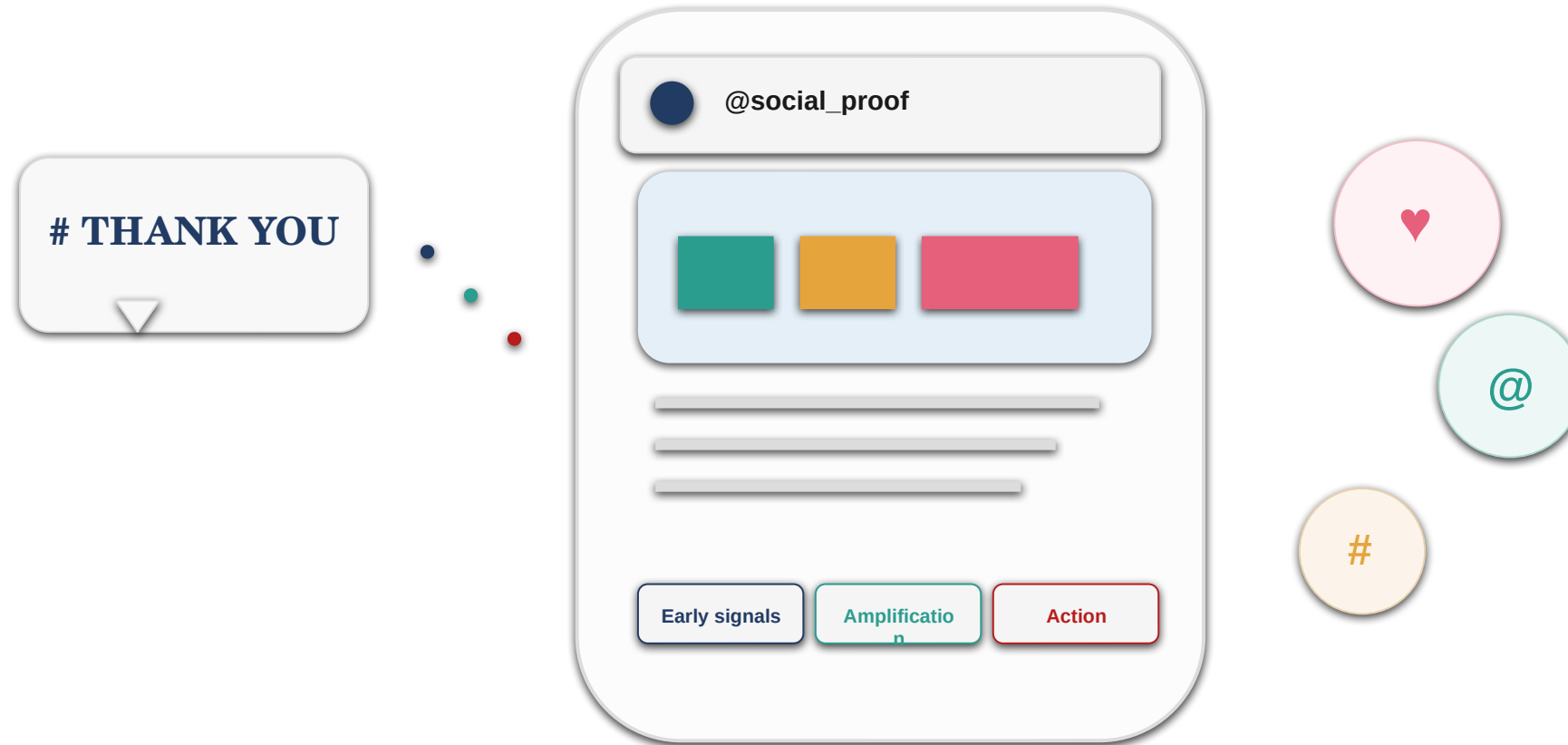
Observational data, proxy amplification measures, and prepared instructional data. Future work should use platform-partnered or experimental data.

Observational design limits causal claims.

Future work should test live multi-platform data.

Final takeaway

On algorithmic platforms, being widely seen is not the same as driving meaningful action.



From Social Proof to Action — thank you for your time.

BACKUP SLIDES

Use only during Q&A. Do not present these during the 5-minute talk unless asked.

Backup A: Detailed Hypotheses / Research Questions

Keep details here so the main talk stays under 5 minutes

Hypot hesis	Statement	Result	Evidence
H1	Early persuasive signals are positively associated with algorithmic amplification.	Supported	coef=0.3551, p=0.0000
H2	Algorithmic amplification mediates the relationship between early persuasive signals and high-effort engagement outcomes.	Supported	indirect_effect=0.0041
H3	Influencer authority positively moderates the relationship between early persuasive signals and algorithmic amplification.	Supported	coef=0.2103, p=0.0000
H4	The indirect effect of early persuasive signals on high-effort engagement through algorithmic amplification is stronger for high-authority influencers.	Supported	low=-0.0105, high=0.0541
H5	Reach is positively associated with high-effort engagement at low-to-moderate levels of visibility.	Supported	coef=0.7436, p=0.0000
H6	The positive effect of reach on high-effort engagement exhibits diminishing marginal returns at higher levels of reach. e	Supported	coef=-0.0262, p=0.0048
H7	Persuasive power is positively associated with high-effort engagement under low-to-moderate visibility conditions.	Supported	coef=0.0304, p=0.0210
H8	Under very high visibility conditions, persuasive power is less predictive of high-effort engagement than under lower visibility conditions.	Supported	coef=-0.3397, p=0.0000

Hypothesis	Predictor / Focal Variable	Outcome Variable	Model Role	Expected Relationship
H1	Early persuasive signals	Algorithmic amplification	Direct effect	Positive association
H2	Early persuasive signals	High-effort engagement	Mediation through algorithmic amplification	Positive indirect association
H3	Early persuasive signals × Influencer authority	Algorithmic amplification	Moderation	Positive moderation; stronger association at higher authority
H4	Early persuasive signals	High-effort engagement	Moderated mediation through amplification, conditioned by authority	Stronger indirect association for high-authority influencers
H5	Reach	High-effort engagement	Direct effect under lower to moderate visibility	Positive association
H6	Reach	High-effort engagement	Nonlinear effect / saturation	Positive association weakens at higher reach
H7	Persuasive power	High-effort engagement	Direct effect under low to moderate visibility	Positive association
H8	Very high visibility × persuasive power	High-effort engagement	Boundary condition / interaction effect	Negative interaction; persuasive power is less predictive under very high visibility

Q&A use: H2 is not supported as a uniform direct effect, but conditional indirect effects show the pathway is meaningful mainly at high authority.

Backup B: Variable Definitions and Feature Engineering

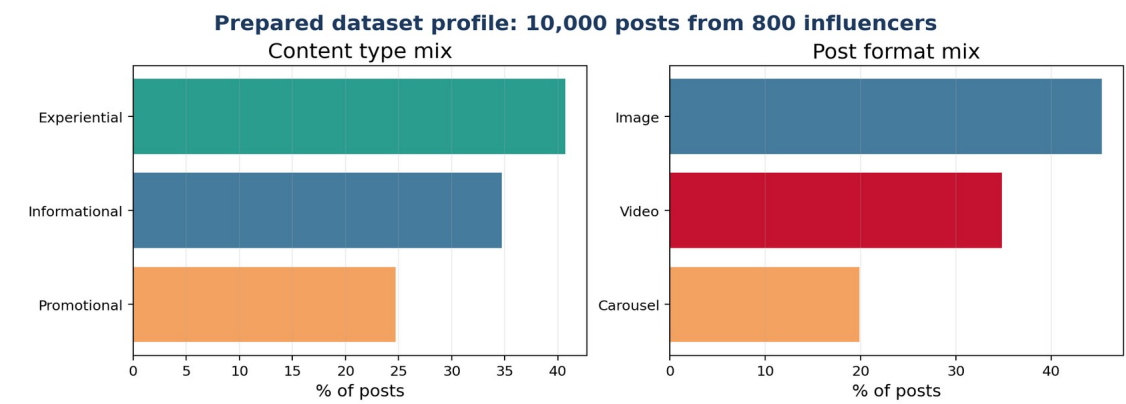
Define constructed indices, transformations, tiers, and formulas

Construct	Operationalization	Why it matters
Early persuasive signals	first-hour engagement; persuasive_power_index	Initial social proof before broader exposure
Algorithmic amplification	reach, impressions, log_reach; index where applicable	Visibility proxy; platform internals not directly observed
High-effort engagement	saves, link_clicks, profile_visits, engagement indices	Deeper behavioral response beyond likes
Influencer authority	follower_count, authority_log, authority tiers	Source credibility and distribution boundary conditions
Reach saturation	log_reach, log_reach ² , very_high_reach	Diminishing returns and high-visibility regimes
Controls	content_type, post_format, verified, account age, time c	Reduces confounding in post-level comparisons

Measurement caution: amplification is inferred from visibility-related proxies, not directly observed recommendation logic.

Backup C: Descriptive Statistics and Data Quality

Sample profile, data quality, and filtering decisions



Variable	Mean	SD	Min	Max
Log Reach	9.287	0.875	6.629	15.425
Log High-Effort Engagement	1.184	0.863	0.000	5.765
Persuasive Power	0.000	0.643	-1.113	4.849
Influencer Authority (Log)	10.315	0.980	7.059	14.153

Data quality decisions

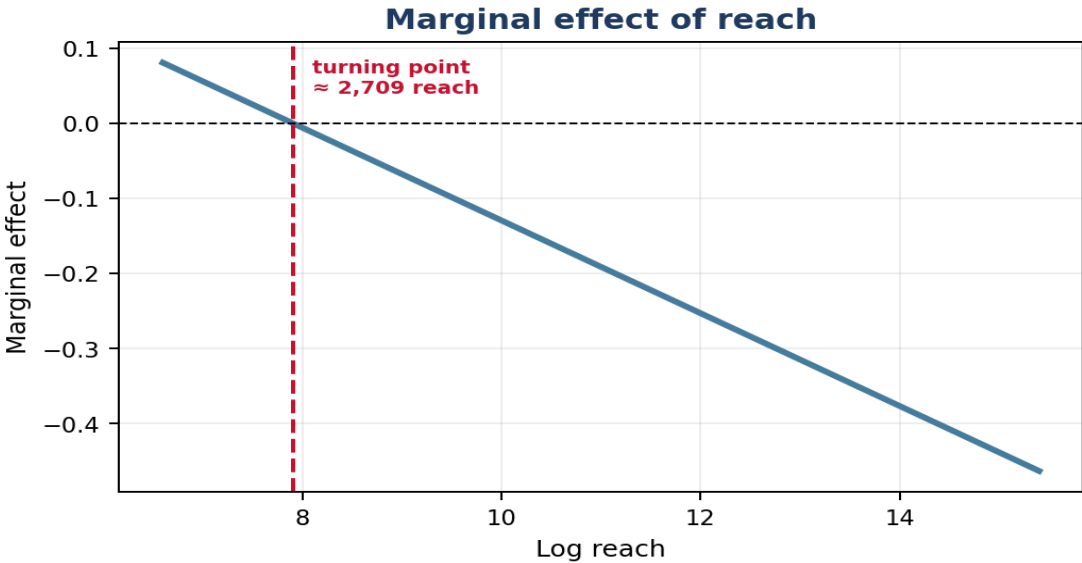
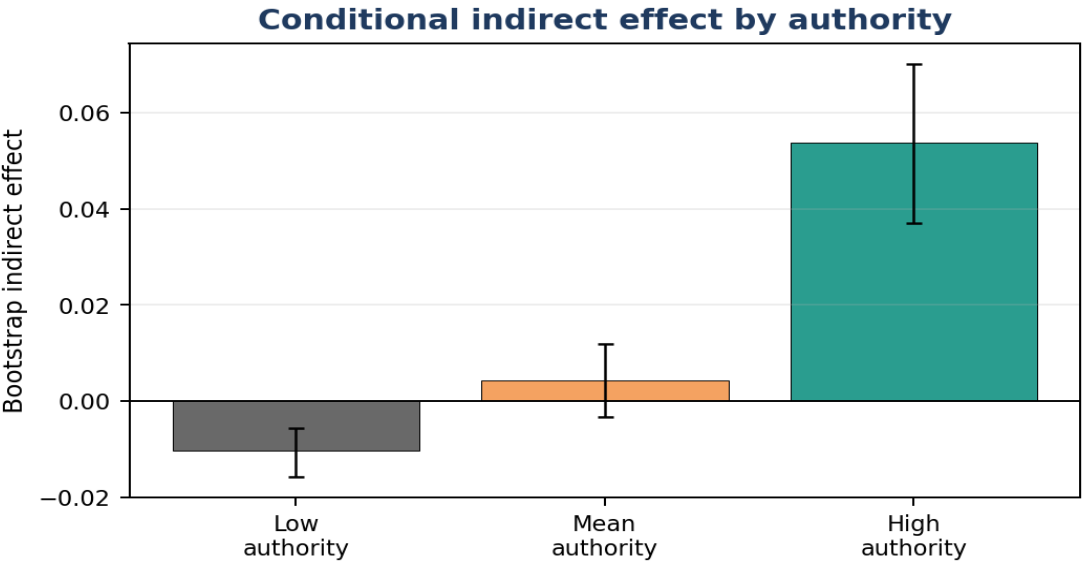
- log transforms for skewed count variables
- casewise exclusion only when focal variables are missing
- clustered standard errors by influencer

Generalizability

The dataset is prepared for analysis and should be interpreted as analytically informative, not a probability sample of all social media contexts.

Backup D: Full Model Results / Robustness Checks

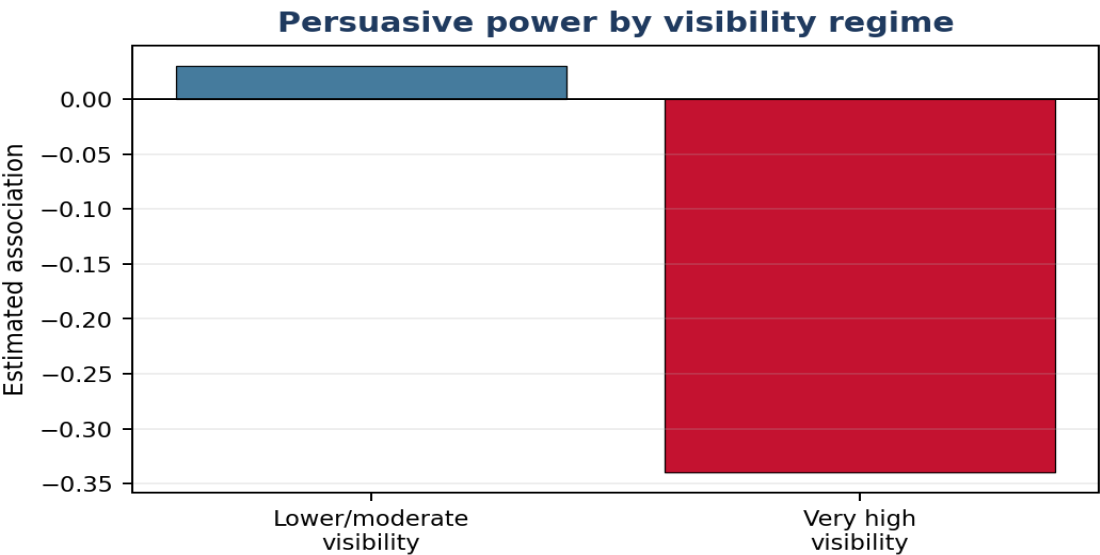
Keep detailed results here for committee questions



Model	Key coefficient	Interpretation
RQ1: a-path	$\beta = 0.3551, p < .001$	Early persuasive signals associate with amplification
RQ1: b-path	$\beta = 0.0116, p = .297$	Direct amplification link is not supported
RQ2: reach ²	$\beta = -0.0309, p < .001$	Diminishing marginal association
RQ3: log reach × persuasive power	$\beta = -0.0389, p < .001$	Persuasive power weaker at high visibility

Backup E: Managerial Recommendations

Additional high-visibility visualization and action guidance



Recommendation 1

Separate reach from effectiveness. Report high-effort outcomes alongside reach and impressions.

Recommendation 2

Invest in early content quality and launch timing because early signals are linked to visibility.

Recommendation 3

Use influencer authority strategically; high-authority creators convert early signals into meaningful outcomes more effectively.

Do not over-optimize for maximum reach. The model suggests diminishing returns and high-visibility boundary effects.

Backup F: References and AI-Use Statement

Abbreviated references and disclosure

- Chaiken (1980) — Heuristic versus systematic information processing.
- Bucher (2012) — Algorithmic power and visibility on Facebook.
- Cotter (2019) — Influencers and algorithmic visibility on Instagram.
- Petre, Duffy, & Hund (2019) — Politics of algorithmic visibility.
- De Veirman, Cauberghe, & Hudders (2017) — Followers and influencer effects.
- Lou & Yuan (2019) — Message value, credibility, and branded content trust.
- Xiao, Wang, & Chan-Olmsted (2018) — HSM and influencer credibility.

AI-use statement

AI tools were used to support drafting, organization, and slide formatting. The analysis and interpretation are based on the provided dataset, code outputs, and project materials.

Final reminder: claims should be framed as associations because the design is observational and amplification is measured through proxies.